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K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai.
Somaiya TRUST

DEPARTMENT OF COMPUTER ENGINEERING

Academic Year: 2026-27(Odd Sem)

Lab Manual

Class –SY Sem - III

Course Name: Skill Based Learning: Object Oriented Programming with Java (SAT-III)

Course Code: CEXS33

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Subject: Skill Based Learning: Object Oriented Programming with Java (SAT-III) **Semester**: III

Course code: UCEXS33 Practical/week: 2 hrs/batch Session: 2026-27

Skill based learning Educational Objectives (SEOs):

Sr. No	Objectives
SEO1	To learn the basic concepts of object-oriented programming
SEO2	To study JAVA programming language
SEO3	To study various concepts of JAVA programming like multithreading, exception Handling, packages, etc.
SEO4	To explain components of GUI based programming

Skill Outcomes (LOs): Learner will be able to…

Index	Outcomes
SO1	Apply fundamental programming constructs.
SO2	Implement the concept of classes and objects, inheritance and interfaces.
SO3	Implement the concept of strings, arrays and packages
SO4	Implement the concept of exception handling and multithreading.
SO5	Develop GUI based application.
SO6	Apply ethical principles like timeliness and adhere to the rules of the laboratory

Mapping of Skill Objectives with Skill Outcomes:

Skill Objectives	Skill Outcomes					
	SO1	SO2	SO3	SO4	SO5	SO6
SEO 1	√	√				√
SEO 2	√	√	√	√	√	√
SEO 3			√	√		√
SEO 4					√	√

Mapping of SOs with POs and PSOs:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
SO1	3	1	1	1	1	1	-	-	-	-	3	1	3	-	2
SO2	3	-	2	1	2	1	-	-	-	-	3	1	3	-	-
SO3	3	-	2	1	2	1	-	-	-	-	3	1	3	-	-
SO4	3	-	2	1	2	1	-	-	-	-	3	1	3	-	-
SO5	3	-	2	1	2	1	-	-	-	-	3	1	3	-	-
SO6	3	3	3	3	3	3	2	2	2	2	3	1	3	-	2

List of Experiments:

Exp.	Name of Experiments	SEOs	SOs
0	Prerequisite lab		
1	Write a program to implement basic programming constructs like branching and looping	SEO1,SEO2	SO1, SO6
2	Write a program to demonstrate different ways of accepting user input in Java	SEO1,SEO2	SO1, SO6
3	Write a program to implement the concept of : 1. Method overloading 2. Constructor overloading	SEO1,SEO2	SO2, SO6
4	Write a program implement the concept of 2D array and String Manipulation functions in Java	SEO2	SO3, SO6
5	Write a program to implement the concept of Inheritance	SEO1,SEO2	SO2, SO6
6	Write a program to implement the concept of Method Overriding	SEO1,SEO2	SO2, SO6
7	Write a program to implement the concept of abstract class and abstract method	SEO1,SEO2	SO2, SO6
8	Write a program to implement the concept of package	SEO2, SEO3	SO3, SO6
9	Write a program to implement the concept of Exception handling	SEO2, SEO3	SO4, SO6
10	Write a program to implement the concept of Multithreading Algorithm	SEO2, SEO3	SO4, SO6
11	Design form for Admission process management application system using AWT or Java Swing	SEO2, SEO4	SO5, SO6

12	Study and Implement the concept of JDBC and Perform CRUD Operation on the form created in 11 using Java Database Connectivity	SEO2, SEO4	SO5, SO6

Experiment No:1

Aim: Write a programs on Basic programming constructs like branching and looping.

A: Write a program implement different arithmetic operations using switch case statement

B. Write a program to display Fibonacci Series by using loop. Also write a program to implement for each loop to display array elements.

Software Required: Jdk1.8.0

Theory:

Control statements in java is one of the fundamentals required for java programming.
It allows the smooth flow of a program.

Java control statements:

Control statements can be divided into three categories, namely

- Selection statements/decision making statements
- Iteration statements
- Jump statements

1.Selection statements/decision making statements

Simple if statement

The if statement determines whether a code should be executed based on the specified condition. Syntax:

```
If (condition) {
Statement 1; //executed if condition is true
 }
Statement 2; //executed irrespective of the condition
```


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If..else statement

In this statement, if the condition specified is true, the if block is executed. Otherwise, the else block is executed.

Nested if statement

An if present inside an if block is known as a nested if block. It is similar to an if..else statement, except they are defined inside another if..else statement.

```
if (condition1) {
Statement 1; //executed if first condition is true
if (condition2) {
Statement 2; //executed if second condition is true
 }
else {
```

Statement 3; //executed if second condition is false

}

}

Switch statement

A switch statement in java is used to execute a single statement from multiple conditions. The switch statement can be used with short, byte, int, long, enum types, etc. Certain points must be noted while using the switch statement: α one or n number of case values can be specified for a switch expression. A case values that are duplicate are not permissible. A compile-time error is generated by the compiler if unique values are not used. A the case value must be literal or constant. Variables are not permissible. A usage of break statement is made to terminate the statement sequence. It is optional to use this statement. If this statement is not specified, the next case is executed.

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2. Looping statements

Statements that execute a block of code repeatedly until a specified condition is met are known as looping statements. Java provides the user with three types of loops:

While loop :

Known as the most common loop, the while loop evaluates a certain condition. If the condition is true, the code is executed. This process is continued until the specified condition turns out to be false.The condition to be specified in the while loop must be a boolean expression. An error will be generated if the type used is int or a string.

while (condition)

{

Statement one;

}

Do….while loop :

The do-while loop is similar to the while loop, the only difference being that the condition in the do-while loop is evaluated after the execution of the loop body. This guarantees that the loop is executed at least once

do{

//code to be executed

}while(condition);

For

The for loop in java is used to iterate and evaluate a code multiple times. When the number of iterations is known by the user, it is recommended to use the for loop.

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Syntax:

for (initialization; condition; increment/decrement)

{Statement;}

for-each

The traversal of elements in an array can be done by the for-each loop. The elements present in the array are returned one by one. It must be noted that the user does not have to increment the value in the for-each loop.

3.Branching statements :

Branching statements in java are used to jump from a statement to another statement, thereby the transferring the flow of execution.

Break :

The break statement in java is used to terminate a loop and break the current flow of the program.

Continue :

To jump to the next iteration of the loop, we make use of the continue statement. This statement continues the current flow of the program and skips a part of the code at the specified condition.

Source Code: Source code of program

Output: Screenshots of output.

Conclusion: Control statements in java are studied. switch case control statement is used to perform different arithmetic operations and looping statements used to display Fibonacci series number and print array elements.

Questions: 1.What are different control statements in Java?

2.What is the difference between while and dowhile,for and foreach ,break and continue.

Experiment No:2

Aim: Write a program to demonstrate different ways of accepting user input in java A: Write a program to calculate area of circle. (Scanner class) B. Write a program to find the number is prime number or not. (BufferedReaderClass)

C. Write a program to display addition of two numbers by using Command Line argument. (Commend Line argument)

Software Required: Jdk1.8.0

Theory:

There are different ways of accepting user input in java.

1.Command Line Argument

2.Scanner class

3.BufferedReader class

1.Command Line Argument

There may be occasions when we may like our program to act in a particular way depending on the input provided at the time of execution. This is achieved by java programs by using what is known as Command Line arguments. They are parameters that are supplied to the application program at the time of invoking at for execution.

Command line arguments are passed in an array of Strings,which is passed as parameter to main() method in any application program. The first array component starts with zero.

2. The Java Scanner class is used to collect user input. Scanner is part of the java.util package, so it can be imported without downloading any external libraries.

In order to work with the Scanner class, you must first import it into your code. There are two ways you can do this:

If you only need to work with the `java.util.Scanner` class, you can import the Scanner class directly.

If you are working with other modules in the the full `java.util` library, you may want to import library.

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The following is the code for each of the above methods:

import java.util.Scanner; // This will import the Scanner

class.

import java.util.*; // This will import the full java.util library—every class within java.util.

Methods of Scanner Class:

Method	Type of Value the Method Collects
nextBoolean()	Boolean
nextByte()	Byte
nextDouble()	Double
nextFloat()	Float
nextInt()	Int
nextLine()	String
nextLong()	Long
nextShort()	Short

Java BufferedReader Class :

Java BufferedReader class is used to read the text from a character-based input stream. It can be used to read data line by line by readLine() method. It makes the performance fast. It inherits Reader class.

Sourse Code: Source code of program



Output: Screenshots of output.

Conclusion: Java provides different ways of accepting user input. Scanner class, Buffered Reader class and Command Line argument approach is used here to demonstrate the same.

Viva Questions: 1.What is Command Line argument

2.How to use BufferedReader Class

3.What is the difference between Scanner class and BufferedReader

Class 4.What are the different methods of Scanner Class to accept

input

Experiment No:3

Aim: Write a program to implement the concept of metho overloading and Constructor overloading

Software Required: Jdk1.8.0

Theory:

Method Overloading is a feature that allows a class to have more than one method having the same name, if their argument lists are different. Method overloading is an example of Static Polymorphism. Static Polymorphism is also known as compile time binding or early binding. Static binding happens at compile time. Method overloading is an example of static binding where binding of method call to its definition happens at Compile time.

Three ways to overload a method

In order to overload a method, the argument lists of the methods must differ in either of these: 1.Number of parameters.
For example: This is a valid case of overloading

add(int, int)

add(int, int, int)

2. Data type of parameters.
For example:

add(int, int)

add(int, float)

3. Sequence of Data type of parameters.
For example:

add(int, float)

add(float, int)

Invalid case of method overloading:
It is not possible

int add(int, int)



float add(int, int)

Constructor Overloading

Constructor overloading is a technique in Java in which a class can have any number of constructors that differ in parameter list. The compiler differentiates these constructors by taking into account the number of parameters in the list and their type.

Examples of valid constructors for class Account are

Account(int a);
Account (int a,int b);
Account (String a,int b);

Sourse Code: Source code of program

Conclusion: The concept of method overloading is studied and implemented to calculate are of different shapes like rectangle, square and circle.

Viva Questions:

1. What do you mean by method overloading?

2. What do you mean by constructor overloading?

3. what do you mean by compile time polymorphism and how to achieve this.



Experiment No: 4

Aim: A. Write a program to perform addition of two matrices using 2D array. B. Write a program to implement different String Manipulation functions.

Software Required: Jdk1.8.0

Theory:

Java array is an object which contains elements of a similar data type. Additionally, The elements of an array are stored in a contiguous memory location. It is a data structure where we store similar elements. We can store only a fixed set of elements in a Java array.



Array in Java is index-based, the first element of the array is stored at the 0th index, 2nd element is stored on 1st index and so on

Advantages

- **Code Optimization:** It makes the code optimized, we can retrieve or sort the data efficiently.
- **Random access:** We can get any data located at an index position.

Disadvantages

- **Size Limit:** We can store only the fixed size of elements in the array. It doesn't grow its size at runtime. To solve this problem, collection framework is used in Java which grows automatically.

There are two types of array.

- Single Dimensional Array



- Multidimensional Array

Syntax to Declare an Array in Java

1. dataType[] arr; (or)
2. dataType []arr; (or)
3. dataType arr[];

Instantiation of an Array in Java

arrayRefVar=**new** datatype[size];

String Class

In Java, a string is a sequence of characters. For example, "hello" is a string containing a sequence of characters 'h', 'e', 'l', 'l', and 'o'.

String Manipulation Function:

Methods Description

concat() joins the two strings together

equals() compares the value of two strings

charAt() returns the character present in the specified location

getBytes() converts the string to an array of bytes

indexOf() returns the position of the specified character in the string

length() returns the size of the specified string

replace() replaces the specified old character with the specified new character

substring() returns the substring of the string

split() breaks the string into an array of strings

toLowerCase() converts the string to lowercase

toUpperCase() converts the string to uppercase



valueOf() returns the string representation of the specified data

Sourse Code: Source code of program

Output: Screenshots of output.

Conclusion: The concept of array and String is studied and

implemented. Viva Questions:

1. What is array? What are different methods of Array?

2. What are different string manipulation functions?



Experiment No:5

Aim: Write a program to implement the concept of Inheritance

A. Write a program to implement multilevel inheritance.

Software Required: Jdk1.8.0

Theory:

Inheritance in Java is a mechanism in which one object acquires all the properties and behaviors of a parent object. It is an important part of OOPs (Object Oriented programming system).

The idea behind inheritance in Java is that you can create new classes that are built upon

existing classes. When you inherit from an existing class, you can reuse methods and fields of the parent class. Moreover, you can add new methods and fields in your current class also.

Inheritance represents the IS-A relationship which is also known as a *parent-child* relationship. **is-a relationship**

Inheritance is an is-a relationship. We use inheritance only if an is-a relationship is present between the two classes.

Here are some examples:

- A car is a vehicle.
- Orange is a fruit.

Why use inheritance in java

- For Method Overriding (so runtime polymorphism can be achieved).
- For Code Reusability.

Terms used in Inheritance

- Class: A class is a group of objects which have common properties. It is a template or blueprint from which objects are created.

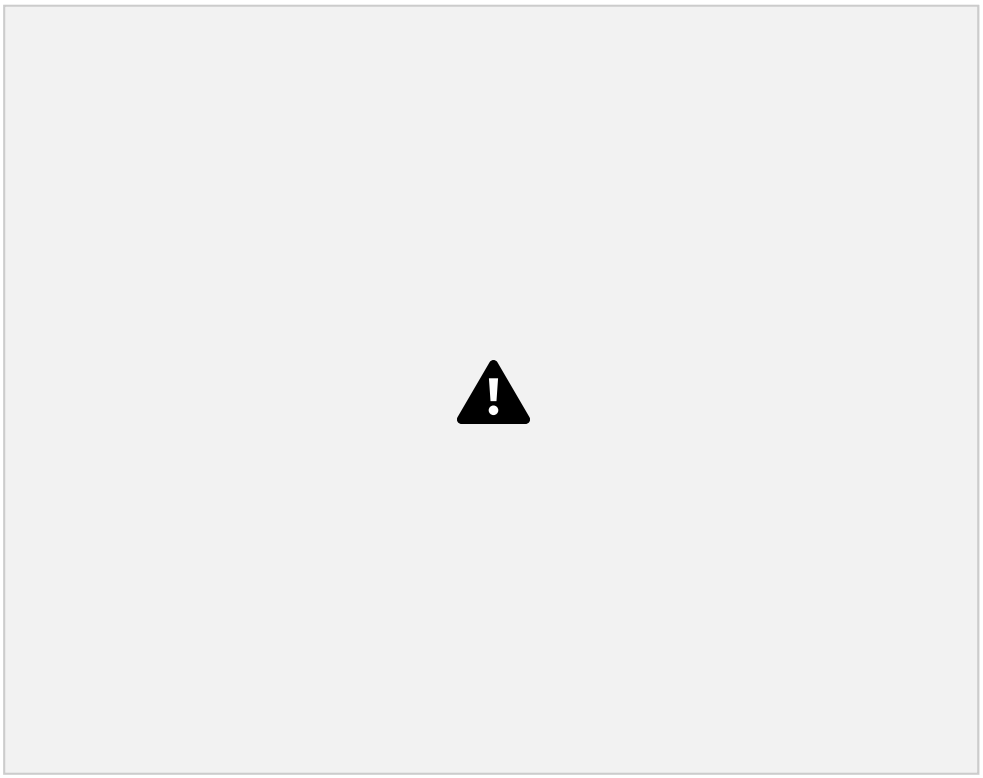


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- Sub Class/Child Class: Subclass is a class which inherits the other class. It is also called a derived class, extended class, or child class.
- Super Class/Parent Class: Superclass is the class from where a subclass inherits the features. It is also called a base class or a parent class.
- Reusability: As the name specifies, reusability is a mechanism which facilitates you to reuse the fields and methods of the existing class when you create a new class. You

can use the same fields and methods already defined in the previous class.

Types of Inheritance:



Sourse Code: Source code of program

Output: Screenshots of output.



Conclusion:

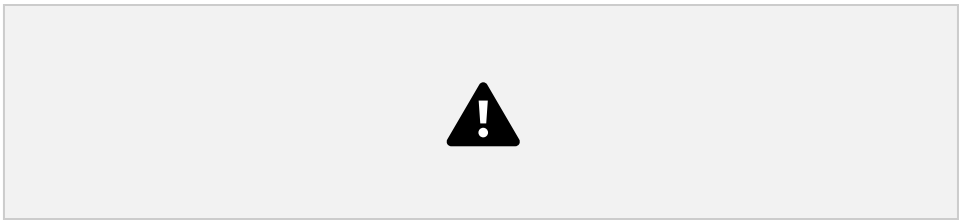
The concept of inheritance and its types is studied and implemented to see the real life usage.

Viva Questions:

1. What is inheritance?

2. Different types of inheritance in Java?

3.Which Keyword is used to achieve inheritance?



Experiment No:6

Aim: Write a program to implement the concept of Method Overriding.

Software Required: Jdk1.8.0

Theory:

Declaring a method in **sub class** which is already present in **parent class** is known as method overriding. Overriding is done so that a child class can give its own implementation to a method which is already provided by the parent class. In this case the method in parent class is called overridden method and the method in child class is called overriding method.

Advantage of method overriding

The main advantage of method overriding is that the class can give its own specific implementation to a inherited method **without even modifying the parent class code**.

This is helpful when a class has several child classes, so if a child class needs to use the parent class method, it can use it and the other classes that want to have different implementation can use overriding feature to make changes without touching the parent class code.

Rules of method overriding in Java

1. Argument list: The argument list of overriding method (method of child class) must match the Overridden method(the method of parent class). The data types of the arguments and their sequence should exactly match.
2. Access Modifier of the overriding method (method of subclass) cannot be more restrictive than the overridden method of parent class. For e.g. if the Access Modifier of parent class method is public then the overriding method (child class method) cannot have private, protected and default Access modifier, because all of these three access modifiers are more restrictive than public.

1. private, static and final methods cannot be overridden as they are local to the class. However static methods can be re-declared in the sub class, in this case the sub-class method would act differently and will have nothing to do with the same static method of parent class.



2. Overriding method (method of child class) can throw unchecked exceptions, regardless of whether the overridden method(method of parent class) throws any exception or

not. However the overriding method should not throw checked exceptions that are new or broader than the ones declared by the overridden method.

3. Binding of overridden methods happen at runtime which is known as dynamic binding.
4. If a class is extending an abstract class or implementing an interface then it has to override all the abstract methods unless the class itself is a abstract class.

Sourse Code: Source code of program

Output: Screenshots of output.

Conclusion: The concept of Method Overriding is studied and

implemented. Viva Questions:

1. What is method overriding?

2. What is use of super keyword?

3. What is the use of final keyword?



Experiment No:7

Aim: A. Write a program to implement the concept of abstract class and abstract method. B. Write a program to demonstrate multiple inheritance using interface.

Software Required: Jdk1.8.0

Theory:

Abstraction is a process of hiding the implementation details and showing only functionality to the user. It shows only essential things to the user and hides the internal details, for example, sending SMS where you type the text and send the message. You don't know the internal processing about the message delivery.

Ways to achieve Abstraction

There are two ways to achieve abstraction in java

1. Abstract class (0 to 100%)

2. Interface (100%)

Abstract class in Java

A class which is declared as abstract is known as an abstract class. It can have abstract and non abstract methods. It needs to be extended and its method implemented. It cannot be instantiated.

o An abstract class must be declared with an abstract keyword.

o It can have abstract and non-abstract methods.

o It cannot be instantiated.

o It can have constructors and static methods also.



o It can have final methods which will force the subclass not to change the body of the method.

Example of abstract class

```
1. abstract class A{}
```

Abstract Method in Java

A method which is declared as abstract and does not have implementation is known as an abstract method.

Example of abstract method

```
1. abstract void printStatus();//no method body and abstract
```

Sourse Code: Source code of program

Output: Screenshots of output.

Conclusion: Concept of abstract class and abstract method is studied and

implemented. Viva Questions:

1.What is abstraction?

2. How to achieve abstraction?

3.What is the difference between abstraction and interface?



Experiment No:8

Aim: Write a program to implement the concept of package

A: Develop a program which consists of the package named let_me_calculate with a class named calculator and a method name add to add two integer numbers, import let_me_calculate package in another program to add two numbers.

B: Develop a program which consists of the package named myinstitute include class name as Department with one method to display the staff of that department. Develop a program to import this package in java application and call the method define in the package.

Software Required: Jdk1.8.0

Theory:

A package is nothing but a physical folder structure (directories) that contains a group of related classes, interfaces, and sub-packages according to their functionality. It provides a convenient way to organize your work. The Java language has various in-built packages.For example, java.lang, java.util, java.io, and java.net. All these packages are defined as a very clear and systematic packaging mechanism for categorizing and managing.

Advantage of using packages in Java

1. Maintenance: Java packages are used for proper maintenance. If any developer newly joined a company, he can easily reach to files needed.

2. Reusability: We can place the common code in a common folder so that everybody can check that folder and use it whenever needed.

3. Name conflict: Packages help to resolve the naming conflict between the two classes with the same name. Assume that there are two classes with the same name Student.java. Each class will be stored in its own packages such as stdPack1 and stdPack2 without having any conflict of names.

4. Organized: It also helps in organizing the files within our project.



5. Access Protection: A package provides access protection. It can be used to provide visibility control. The members of the class can be defined in such a manner that they will be visible only to elements of that package.

Types of Packages in Java
There are two different types of packages in Java. They are:

1. User-defined Package
The package which is defined by the user is called a User-defined package. It contains user defined classes and interfaces.

Creating package: package packageName;

```
package myPackage;

public class A {

// class body

}
```

2. Predefined Packages in Java (Built-in Packages)

Predefined packages in java are those which are developed by Sun Microsystem. They are also called built-in packages in java. These packages consist of a large number of predefined classes, interfaces, and methods that are used by the programmer to perform any task in his

programs.

Core packages:

1. Java.lang: lang stands for language. The Java language package consists of java classes and interfaces that form the core of the Java language and the JVM. It is a fundamental package which is useful for writing and executing all Java programs.

Examples are classes, objects, String, Thread, predefined data types, etc. It is imported automatically into the Java programs.

2. Java.io: io stands for input and output. It provides a set of I/O streams that are used to read and write data to files. A stream represents a flow of data from one place to another place.



3. Java util: util stands for utility. It contains a collection of useful utility classes and related interfaces that implement data structures like LinkedList, Dictionary, HashTable, stack, vector, Calender, data utility, etc.

4. Java.net: net stands for network. It contains networking classes and interfaces for networking operations. The programming related to client-server can be done by using this package.

Window Toolkit and Applet:

1. Java.awt: awt stands for abstract window toolkit. The Abstract window toolkit packages contain the GUI(Graphical User Interface) elements such as buttons, lists, menus and text areas. Programmers can develop programs with colorful screens, paintings, and images, etc using this package.

2. Java.awt.image: It contains classes and interfaces for creating images and colors.

3. Java.applet: It is used for creating applets. Applets are programs which are executed from the server into the client machine on a network.

4. Java.text: This package contains two important classes such as DateFormat and NumberFormat. The class DateFormat is used to format dates and times. The NumberFormat

is used to format numeric values.

5. Java.sql: SQL stands for structured query language. This package is used in a Java program to connect databases like Oracle or Sybase and retrieve the data from them.

Source Code: Source code of program

Output: Screenshots of output.

Conclusion: The concept of package is studied and

implemented. Viva Questions:

1. What is package?



2. What is the advantage of using package?

3. What is the difference between inbuilt package and user defined package?



Experiment No:9

Aim: Write a program to implement the concept of Exception Handling.

A : Write any java program which illustrates the try, catch and finally blocks. B : Write a program that handles an exception "Entry of negative age of person" using throw and thows.

Software Required: Jdk1.8.0

Theory:

Exception handling

Exception handling is one of the most important feature of java programming that allows us to handle the runtime errors caused by exceptions.

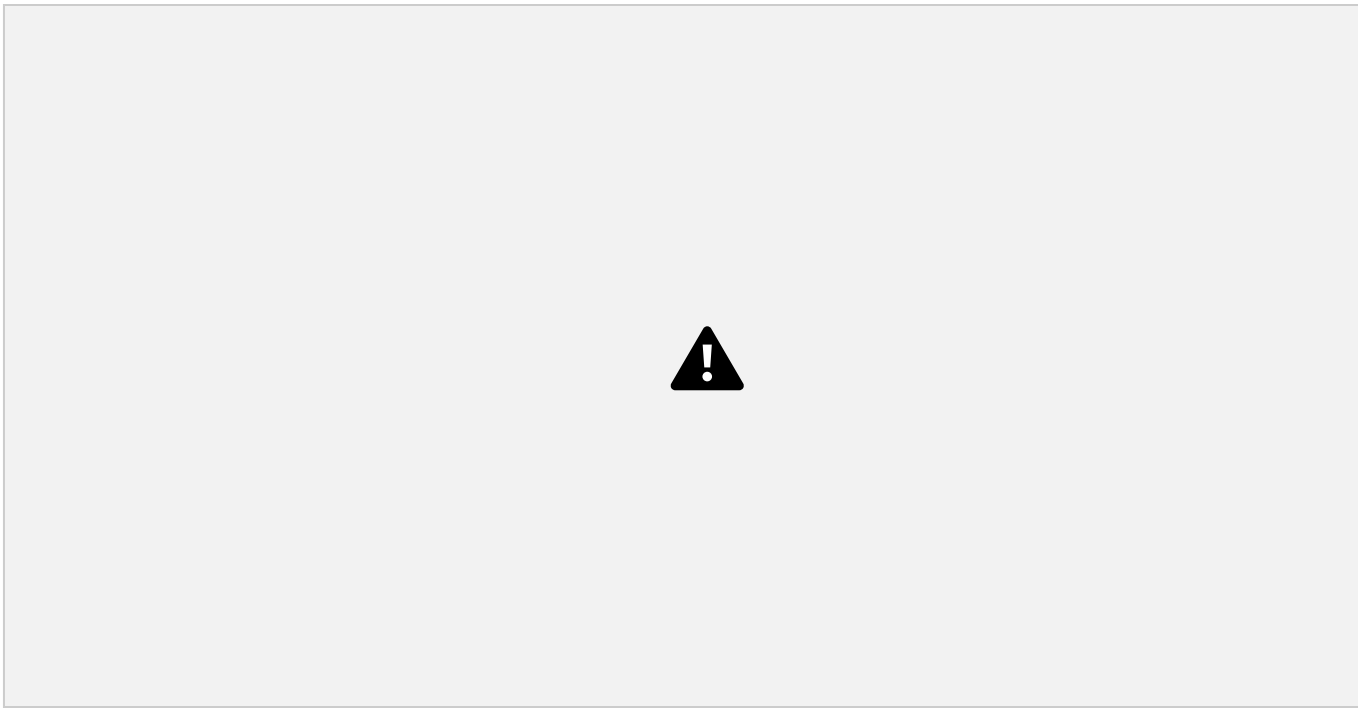
What is an exception?

An Exception is an unwanted event that interrupts the normal flow of the program. When an exception occurs program execution gets terminated. In such cases we get a system generated error message. The good thing about exceptions is that they can be handled in Java. By handling the exceptions we can provide a meaningful message to the user about the issue rather than a system generated message, which may not be understandable to a user.

Why an exception occurs?

There can be several reasons that can cause a program to throw exception. For example: Opening a non-existing file in your program, Network connection problem, bad input data provided by user etc.





Types of exceptions

There are two types of exceptions in Java:

1)Checked exceptions

2)Unchecked exceptions

Checked exceptions

All exceptions other than Runtime Exceptions are known as Checked exceptions as the compiler checks them during compilation to see whether the programmer has handled them or not. If these exceptions are not handled/declared in the program, you will get compilation error. For example, SQLException, IOException, ClassNotFoundException etc.

Unchecked Exceptions

Runtime Exceptions are also known as Unchecked Exceptions. These xceptions are not checked at compile-time so compiler does not check whether the programmer has handled them or not but it’s the responsibility of the programmer to handle these exceptions and provide a safe exit. For example, ArithmeticException, NullPointerException, ArrayIndexOutOfBoundsException etc.



Java provides five keyword for Exception Handling :

1. try
2. catch
3. finally
4. throw
5. throws

• try-catch is a block of statements to try some commands and trap the runtime errors. • A try can have many catch statements.

• finally is again a block to always execute some code irrespective of an exception. A try can have only one finally block at the bottom.

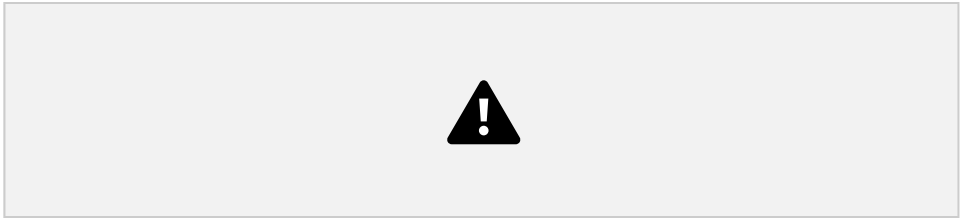
• throw keyword is used to throw an object of some kind of class exception and • throws keyword is used to carry forward that exception object. Sourse Code: Source code of program

Output: Screenshots of output.

Conclusion: The concept of exception handling and usage of keywords like try,catch,throw,throws and finally is studied and implemented.

Viva Questions:

1. What is Exception?
2. Why exception handling is important?
3. What is the usage of keywords like try,catch,throw,throws and finally?



Experiment No:10

Aim: Write a program to implement the concept of Multithreading.
Software Required: Jdk1.8.0

1.Create a Thread by Extending a Thread Class

Theory:

Multithreading is a Java feature that allows concurrent execution of two or more parts of a program for maximum utilization of CPU. Each part of such program is called a thread. So, threads are light-weight processes within a process.Java is a multi-threaded programming language which means we can develop multi-threaded program using Java. A multi-threaded program contains two or more parts that can run concurrently and each part can handle a different task at the same time making optimal use of the available resources specially when your computer has multiple CPUs.Multi-threading enables you to write in a way where multiple activities can proceed concurrently in the same program.

Ways to Create Thread:

2.Create a Thread by Implementing a Runnable Interface

Thread Methods

Following is the list of important methods available in the Thread class.

Sr.No.	Method & Description
1	**public void start()** Starts the thread in a separate path of execution, then invokes the run() method on this Thread object.

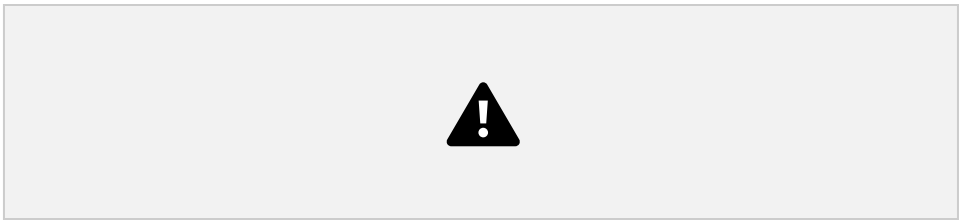


2	**public void run()** If this Thread object was instantiated using a separate Runnable target, the run() method is invoked on that Runnable object.
3	**public final void setName(String name)** Changes the name of the Thread object. There is also a getName() method for retrieving the name
4	**public final void setPriority(int priority)** Sets the priority of this Thread object. The possible values are between 1 and 10.
5	**public final void setDaemon(boolean on)** A parameter of true denotes this Thread as a daemon thread.
6	**public final void join(long millisec)** The current thread invokes this method on a second thread, causing the current thread to block until the second thread terminates or the specified number of milliseconds passes.
7	**public void interrupt()** Interrupts this thread, causing it to continue execution if it was blocked for any reason.
8	**public final boolean isAlive()** Returns true if the thread is alive, which is any time after the thread has been started but before it runs to completion.

Sourse Code: Source code of program

Output: Screenshots of output.

Conclusion: The concept of multithreading is studied and



implemented. Viva Questions:

1. What is Multithreading?

2. Explain Thread Life cycle?

3. What are the different ways to create Thread?



Experiment No:11

Aim: Design form for Admission process management application system using AWT or Java Swing
Software Required: Jdk1.8.0

Theory:

Java Applet

Applet is a special type of program that is embedded in the webpage to generate the dynamic content. It runs inside the browser and works at client side.

Advantage of Applet

There are many advantages of applet. They are as follows:

o It works at client side so less response time.

o Secured

o It can be executed by browsers running under many plateforms, including Linux, Windows, Mac Os etc.

Important points :

1. All applets are sub-classes (either directly or indirectly) of java.applet.Applet class.

2. Applets are not stand-alone programs. Instead, they run within either a web browser or an applet viewer. JDK provides a standard applet viewer tool called applet viewer.

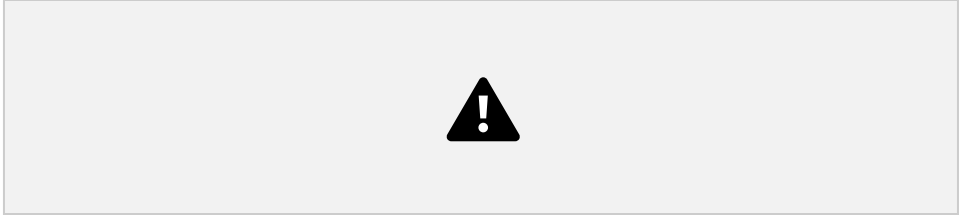
3. In general, execution of an applet does not begin at main() method.

4. Output of an applet window is not performed by System.out.println(). Rather it is handled with various AWT methods, such as drawString().

Lifecycle methods for Applet:

The java.applet.Applet class 4 life cycle methods and java.awt.Component class provides 1 life cycle methods for an applet.

java.applet.Applet class



For creating any applet java.applet.Applet class must be inherited. It provides 4 life cycle methods of applet.

1. public void init(): is used to initialized the Applet. It is invoked only once.

2. public void start(): is invoked after the init() method or browser is maximized. It is used to start the Applet.

3. public void stop(): is used to stop the Applet. It is invoked when Applet is stop or browser is minimized.

4. public void destroy(): is used to destroy the Applet. It is invoked only once.

java.awt.Component class

The Component class provides 1 life cycle method of applet.

1. public void paint(Graphics g): is used to paint the Applet. It provides Graphics class object that can be used for drawing oval, rectangle, arc etc.

Commonly used methods of Graphics class:

1. **public abstract void drawString(String str, int x, int y):** is used to draw the specified string.
2. **public void drawRect(int x, int y, int width, int height):** draws a rectangle with the specified width and height.
3. **public abstract void fillRect(int x, int y, int width, int height):** is used to fill rectangle with the default color and specified width and height.
4. **public abstract void drawOval(int x, int y, int width, int height):** is used to draw oval with the specified width and height.
5. **public abstract void fillOval(int x, int y, int width, int height):** is used to fill oval with the default color and specified width and height.
6. **public abstract void drawLine(int x1, int y1, int x2, int y2):** is used to draw line between the points(x1, y1) and (x2, y2).
7. **public abstract boolean drawImage(Image img, int x, int y, ImageObserver observer):** is used draw the specified image.
8. **public abstract void drawArc(int x, int y, int width, int height, int startAngle, int arcAngle):** is used draw a circular or elliptical arc.
9. **public abstract void fillArc(int x, int y, int width, int height, int startAngle, int arcAngle):** is used to fill a circular or elliptical arc. 10. **public abstract void setColor(Color c):** is used to set the graphics current color to the specified color.
11. **public abstract void setFont(Font font):** is used to set the graphics current font to the specified font.



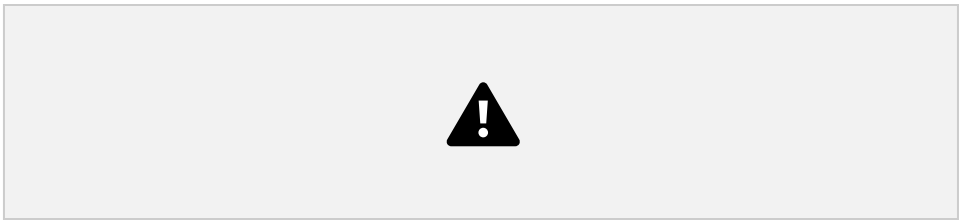
Sourse Code: Source code of program

Output: Screenshots of output.

Conclusion: The concept of applet and Graphics is used to display smiley

face Questions:

1. What is applet?
2. what is the difference between applet and application?
3. Explain applet life cycle?



Experiment No:12

Aim: Study and Implement the concept of JDBC and Perform CRUD Operation using Java Database Connectivity (Select/Update/Delete operation) using Virtual Lab

Link: http://vlabs.iitb.ac.in/vlabs

dev/vlab_bootcamp/bootcamp/bots_with_dots/labs/index.html http://vlabs.iitb.ac.in/vlabs

dev/vlab_bootcamp/bootcamp/bots_with_dots/labs/exp1/

Software Required: Jdk1.8.0

Theory:

What is JDBC ?

JDBC is an acronym for Java Database Connectivity. It's an advancement for ODBC (Open Database Connectivity). JDBC is an standard API specification developed in order to move data from frontend to backend. This API consists of classes and interfaces written in Java

Steps for connectivity between Java program and database:

1. Loading the Driver

To begin with, you first need load the driver or register it before using it in the program . Registration is to be done once in your program. You can register a driver in one of two ways mentioned below :

Class.forName() : Here we load the driver's class file into memory at the runtime. No need of using new or creation of object .The following example uses Class.forName() to load the Oracle driver –

Class.forName(―oracle.jdbc.driver.OracleDriver‖);

DriverManager.registerDriver(): DriverManager is a Java inbuilt class with a static member register. Here we call the constructor of the driver class at compile time . The following example uses DriverManager.registerDriver()to register the Oracle driver –

DriverManager.registerDriver(new oracle.jdbc.driver.OracleDriver())



2. Create the connections

After loading the driver, establish connections using :

Connection con = DriverManager.getConnection(url,user,password)

user – username from which your sql command prompt can be accessed.

password – password from which your sql command prompt can be

accessed. con: is a reference to Connection interface.

url : Uniform Resource Locator. It can be created as follows:

String url = ― jdbc:oracle:thin:@localhost:1521:xe‖

Where oracle is the database used, thin is the driver used , @localhost is the IP Address where database is stored, 1521 is the port number and xe is the service provider. All 3 parameters above are of String type and are to be declared by programmer before calling the function. Use of this can be referred from final code.

3. Create a statement

Once a connection is established you can interact with the database. The JDBCStatement, CallableStatement, and PreparedStatement interfaces define the methods that enable you to send SQL commands and receive data from your database.

Use of JDBC Statement is as follows:

Statement st = con.createStatement();

Here, con is a reference to Connection interface used in previous step .

4. Execute the query

Now comes the most important part i.e executing the query. Query here is an SQL Query . Now we know we can have multiple types of queries. Some of them are as follows:

Query for updating / inserting table in a database.

Query for retrieving data .

The executeQuery() method of Statement interface is used to execute queries of retrieving



values from the database. This method returns the object of ResultSet that can be used to get all the records of a table.

The executeUpdate(sql query) method ofStatement interface is used to execute queries of updating/inserting .

5.Close the connections

So finally we have sent the data to the specified location and now we are at the verge of completion of our task .

By closing connection, objects of Statement and ResultSet will be closed automatically. The close() method of Connection interface is used to close the connection.

Sourse Code: Source code of program

Output: Screenshots of output.

Conclusion: Different database operations are performed using JDBC .

Questions: 1.What is JDBC?

2. what is How to perform select operation?

3. What is PreparedStatement in Java?